

Exercise Sheet 3: Polynomial Interpolation

H By Hand; **C** Computer; **T** Theory; **E** Extra; **A** Advanced.

Recommended: H1.*; C4; H6.b; H7; H8.a; C9.a; H10.b; C13; C14.*

Taylor Series

H1. Let $f(x) = \sqrt{x}$.

- Compute $f'(x)$, $f''(x)$ and $f'''(x)$.
- Find the quadratic Taylor polynomial p_2 and cubic Taylor polynomial $p_3(x)$ around $x_0 = 4$.
- Use $p_2(x)$ and $p_3(x)$ to approximate $\sqrt{4.5}$, $\sqrt{4.25}$, $\sqrt{3.75}$ and $\sqrt{3.5}$ (without using a calculator).
- How would you expect the errors $|f(x) - p_2(x)|$ and $|f(x) - p_3(x)|$ to depend on $|x - x_0|$?
- Determine the actual error of the approximations in (b), and relate this to your predictions in (c).

Answer

- $f(x) = x^{1/2}$, $f'(x) = \frac{1}{2}x^{-1/2}$, $f''(x) = -\frac{1}{4}x^{-3/2}$, $f'''(x) = \frac{3}{8}x^{-5/2}$.
- For $x_0 = 4$, $f(x_0) = 2$, $f'(x_0) = \frac{1}{4}$, $f''(x_0) = -\frac{1}{32}$, $f'''(x_0) = \frac{3}{8} \frac{1}{\sqrt{4}^5} = \frac{3}{256}$, so $p_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1/32}{2!}(x-4)^2 = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$; $p_3(x) = 2 + \frac{1}{4}(x-4) + \frac{1}{64}(x-4)^2 + \frac{1}{512}(x-4)^3$.
 $p_2(4.5) = 2 + \frac{1}{4}(4.5-4) - \frac{1}{64}(4.5-4)^2 = 2 + \frac{1}{4}0.5 - \frac{1}{64}0.5^2 = 2 + \frac{1}{8} - \frac{1}{256} = 2\frac{31}{256} = 2.12109365 = 2.12109$.
 $p_2(4.25) = 2.0615234375$, $p_2(3.75) = 1.9365234375$, $p_2(3.5) = 1.87109375$. $p_3(4.5) = 2 + \frac{1}{4}(4.5-4) - \frac{1}{64}(4.5-4)^2 + \frac{1}{512}(4.5-4)^3 = 2 + \frac{1}{4}0.5 - \frac{1}{64}0.5^2 + \frac{1}{512}0.5^3 = 2\frac{497}{4096} = 2.121337890625 = 2.121338$ (6dp).
 $p_3(4.25) = 2.061553955078$ (12dp), $p_3(3.75) = 1.936492919922$, $p_3(3.5) = 1.870849609375$.
- Error of p_2 is $\frac{1}{6}|f'''(\xi)||x - x_0|^3$, so $O(|x - x_0|^3)$, and error of p_3 is $O(|x - x_0|^4)$.
- Errors $|f(4.5) - p_2(4.5)| = 2.3 \times 10^{-4}$, $|f(4.25) - p_2(4.25)| = 2.9 \times 10^{-5}$, $|f(3.75) - p_2(3.75)| = 3.2 \times 10^{-5}$, $|f(3.5) - p_2(3.5)| = 2.7 \times 10^{-4}$.
 $|f(4.5) - p_3(4.5)| = 1.8 \times 10^{-5}$, $|f(4.25) - p_3(4.25)| = 1.1 \times 10^{-6}$, $|f(3.75) - p_3(3.75)| = 1.2 \times 10^{-6}$,
 $|f(3.5) - p_3(3.5)| = 2.1 \times 10^{-5}$. Error ratios $|f(4.5) - p_2(4.5)|/|f(4.25) - p_2(4.25)| = 7.7 \approx 8 = 2^3$,
 $|f(4.5) - p_3(4.5)|/|f(4.25) - p_3(4.25)| = 15.4 \approx 16 = 2^4$ as expected.

E2. Use the linear and cubic Taylor polynomials of $\sin(x)$ to approximate $\sin(2^\circ)$. Use the error term of the Taylor polynomial to bound the error of the approximation.

Hint: Use the bounds $|\sin(x)|, |\cos(x)| \leq 1$ for all x .

Answer Note that the cubic and quartic Taylor polynomials of $\sin$ about $x_0 = 0$, are the same, $p_3(x) = p_4(x) = x - \frac{1}{6}x^3$, so we can take the error from the quartic ($n = 4$) case, which is $\frac{1}{5!}f^{(5)}(\xi)x^5$. $2^\circ = \frac{2\pi}{180}\text{rad} = 0.034906585040$ (12dp). Taylor series $\sin(x) = x - \frac{1}{6}x^3 + \frac{1}{120}\cos(\xi)x^5$.

Linear approximation, cubic error $|\sin(x) - x| \leq \frac{1}{6}|\cos(\xi)||x^3| \leq \frac{1}{6}|x|^3$, so $\sin(2^\circ) = 0.0349066 \pm 7.1 \times 10^{-6} = 0.03490[66 \pm 71]$.

Cubic approximation $\sin(x) = x - \frac{1}{6}x^3 \pm \frac{1}{120}|x^5|$, so $\sin(2^\circ) = 0.03489949627 \pm 4.3 \times 10^{-10} = 0.034899496[27 \pm 43]$.

A3. One definition of π is as the first positive solution of $\sin(x) = 0$. Use the bisection method starting with $a = 3$ and $b = 3.5$ and the Taylor series for $\sin$ with remainder term to estimate π to an accuracy of 10^{-2} . Do *not* use the built-in `sin` function!

Hint: At all steps of the bisection method, you should evaluate $\sin(x)$ to an accuracy sufficient to determine whether the value is strictly positive or strictly negative.

Polynomial Interpolation

C4. Use the Matlab command `polyfit` to compute the polynomial interpolating the data:

x	0.0	0.6	1.5	1.7	1.9	2.1	2.3	2.6	2.8	3.0
y	0.13	0.38	1.50	1.49	1.12	0.98	1.15	1.89	2.35	2.28

Check your answer by using the Matlab command `polyval` to evaluate the polynomial at the interpolation points. Estimate the value of y when $x = 1.0$ and $x = 2.0$.

Plot the polynomial over the intervals $[0, 3]$ and over the interval $[1.5, 3]$, and compare with plots of the raw data. Comment on whether your estimates for y are reliable.

Hint: The command `cs=polyfit(xs,ys,n)` to computes the *coefficients* c_i of the interpolating polynomial for $n+1$ data points. You can define the polynomial *function* by `p=@(x)polyval(cs,x)`.

Answer

```
xs=[0.0,0.6,1.5,1.7,1.9,2.1,2.3,2.6,2.8,3.0];
ys=[0.13,0.38,1.50,1.49,1.12,0.98,1.15,1.89,2.35,2.28];
deg=numel(xs)-1;
cs=polyfit(xs,ys,deg);
p=@(x)polyval(cs,x);
plot(xs,ys,'x'); hold on; fplot(p,[0,3])
figure
plot(xs,ys,'x'); hold on; fplot(p,[1.5,3])
```

HC5. Write down and sketch the Lagrange basis polynomials $L_{3,k}$ for the values x below.

x	0.6	1.5	1.7	1.9
y	0.38	1.50	1.49	1.12

Use your answer to compute the interpolating polynomial p for the data, and estimate the value of y when $x = 1.0$.

By defining appropriate y -values, use Matlab's `polyfit` command to compute the Lagrange basis polynomials, and the interpolating polynomial, and plot these on the same axes.

Answer

$$\begin{aligned} L_{3,0} &= \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} = \frac{(x-1.5)(x-1.7)(x-1.9)}{(0.6-1.5)(0.6-1.7)(0.6-1.9)} \\ &= \frac{(x-1.5)(x-1.7)(x-1.9)}{(-0.9) \times (-1.1) \times (-1.3)} = \frac{x^3 - 5.1x + 8.63x - 4.845}{-1.287} \\ &= -0.77700x^3 + 3.96270x^2 - 6.70552x + 3.76457 \end{aligned}$$

$$L_{3,1} = \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} = \frac{(x-0.6)(x-1.7)(x-1.9)}{(1.5-0.6)(1.5-1.7)(1.5-1.9)}$$

$$L_{3,2} = \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} = \frac{(x-0.6)(x-1.5)(x-1.9)}{(1.7-0.6)(1.7-1.5)(1.7-1.9)}$$

$$L_{3,3} = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} = \frac{(x-0.6)(x-1.5)(x-1.7)}{(1.9-0.6)(1.9-1.5)(1.9-1.7)}$$

$$\begin{aligned} p(x) &= y_0 L_{3,0} + y_1 L_{3,1} + y_2 L_{3,2} + y_3 L_{3,3} \\ &= 0.38 \frac{(x-1.5)(x-1.7)(x-1.9)}{(0.6-1.5)(0.6-1.7)(0.6-1.9)} + 1.50 \frac{(x-0.6)(x-1.7)(x-1.9)}{(1.5-0.6)(1.5-1.7)(1.5-1.9)} \\ &\quad + 1.49 \frac{(x-0.6)(x-1.7)(x-1.9)}{(1.5-0.6)(1.5-1.7)(1.5-1.9)} + 1.12 \frac{(x-0.6)(x-1.7)(x-1.9)}{(1.5-0.6)(1.5-1.7)(1.5-1.9)} \\ &= -2.55633x^3 + 8.53730x^2 - 7.71115x + 2.48543 \end{aligned}$$

H6. Write down explicitly the Lagrange basis polynomials for the following sets of interpolation points:

- a.** $x_0 = a, x_1 = b$. **b.** $x_0 = -h, x_1 = 0, x_2 = +h$. **c.** $x_0 = 0, x_1 = h, x_2 = 2h$.

Hence, or otherwise, write down the formula for the polynomial interpolating a function f at the given data points.

Note: This is an important theoretical question which will come in useful for finding differentiation formulae!

Answer

- a.** $L_0 = (x - b)/(a - b), L_1 = (x - a)/(b - a),$

$$p(x) = \frac{x-b}{a-b}f(a) + \frac{x-a}{b-a}f(b) = \frac{(x-a)f(b) - (x-b)f(a)}{b-a} = \frac{f(b) - f(a)}{b-a}x + \frac{bf(a) - af(b)}{b-a}.$$

- b.** $L_0 = x(x-h)/2h^2, L_1 = (x+h)(x-h)/(-h^2) = (h^2 - x^2)/h^2 = 1 - x^2/h^2 = 1 - (x/h)^2,$
 $L_2(x) = (x+h)x/2h^2.$

$$\begin{aligned} p(x) &= \frac{x(x-h)f(-h) - 2(x^2 - h^2)f(0) + x(x+h)f(h)}{2h^2} \\ &= f(0) + \frac{f(h) - f(-h)}{2h}x + \frac{f(-h) - 2f(0) + f(h)}{2h^2}x^2. \end{aligned}$$

- c.** $L_0 = (x-h)(x-2h)/2h^2, L_1 = x(x-2h)/(-h^2) = (2h-x)x/h^2, L_2(x) = x(x-h)/2h^2.$

$$\begin{aligned} p(x) &= \frac{(x-h)(x-2h)f(0) - 2x(x-2h)f(h) + x(x-h)f(2h)}{2h^2} \\ &= f(0) + \frac{-3f(0) + 4f(h) - f(2h)}{2h}x + \frac{f(0) - 2f(h) + f(2h)}{2h^2}x^2. \end{aligned}$$

H7. For $k = 1, 2, 3$, compute the interpolating polynomial of degree k for the data $(x_i, y_i)_{i=0, \dots, k}$ below using (i) Neville's method, (ii) nested form, directly computing the coefficients a_i , and (iii) divided differences.

i	0	1	2	3
x_i	0.6	1.5	1.7	1.9
y_i	0.38	1.50	1.49	1.12

Answer

- (i) Neville's method

$$p_{i,i+1, \dots, i+j}(x) = \frac{(x - x_i)p_{i+1, \dots, i+j}(x) - (x - x_{i+j})p_{i, \dots, i+j-1}(x)}{x_{i+j} - x_i}$$

Degree 0 polynomials (constant — just the data values)

$$\left. \begin{aligned} p_0(x) &= 0.38 \\ p_1(x) &= 1.50 \\ p_2(x) &= 1.49 \\ p_3(x) &= 1.12 \end{aligned} \right\} \text{degree 0}$$

Degree 1 polynomials

$$p_{01}(x) = \frac{(x - x_0)p_1 - (x - x_1)p_0}{x_1 - x_0} = \frac{(x - 0.6)(1.50) - (x - 1.5)(0.38)}{0.9} = \frac{1.12x - 0.33}{0.9}$$

$$p_{12}(x) = \frac{(x - x_1)p_2 - (x - x_2)p_1}{x_2 - x_1} = \dots = -0.05x + 1.575$$

$$p_{23}(x) = \frac{(x - x_2)p_3 - (x - x_3)p_2}{x_3 - x_2} = \dots = -1.85x + 4.635$$

$$\left. \begin{aligned} p_{01}(x) &= \frac{1.12x-0.33}{0.9} \\ p_{12}(x) &= -0.05x + 1.575 \\ p_{23}(x) &= -1.85x + 4.635 \end{aligned} \right\} \text{degree 1}$$

Degree 2 polynomials

$$p_{012}(x) = \frac{(x-x_0)p_{12}(x) - (x-x_2)p_{01}(x)}{x_2-x_0} = \dots = -1.1767x^2 + 3.7156x - 1.4258$$

$$p_{123}(x) = \frac{(x-x_1)p_{23}(x) - (x-x_3)p_{12}(x)}{x_3-x_1} = \dots = -4.5x^2 + 14.35x - 9.9$$

$$\left. \begin{aligned} p_{012}(x) &= -1.1767x^2 + 3.7156x - 1.4258 \\ p_{123}(x) &= -4.5x^2 + 14.35x - 9.9 \end{aligned} \right\} \text{degree 2}$$

Degree 3 polynomial

$$p_{0123}(x) = \frac{(x-x_0)p_{123}(x) - (x-x_3)p_{012}(x)}{x_3-x_0} = \dots$$

$$p_{0123}(x) = -2.5564x^3 + 8.5375x^2 - 7.7112x + 2.4854$$

$$p_{0123}(x) = -2.5564x^3 + 8.5375x^2 - 7.7112x + 2.4854 \quad \} \text{degree 3}$$

(ii) Lagrange Interpolation

Each basis polynomial $L_i(x)$ satisfies $L_i(x_i) = 1$ and $L_i(x_j) = 0$ for $j \neq i$:

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x-x_j}{x_i-x_j}$$

Degree 1 (nodes x_0, x_1)

$$L_0(x) = \frac{x-1.5}{0.6-1.5} = \frac{x-1.5}{-0.9} \quad L_1(x) = \frac{x-0.6}{1.5-0.6} = \frac{x-0.6}{0.9}$$

$$p_1(x) = 0.38 \cdot \frac{x-1.5}{-0.9} + 1.50 \cdot \frac{x-0.6}{0.9}$$

Degree 2 (nodes x_0, x_1, x_2)

$$L_0(x) = \frac{(x-1.5)(x-1.7)}{(0.6-1.5)(0.6-1.7)} = \frac{(x-1.5)(x-1.7)}{(-0.9)(-1.1)} = \frac{(x-1.5)(x-1.7)}{0.99}$$

$$L_1(x) = \frac{(x-0.6)(x-1.7)}{(1.5-0.6)(1.5-1.7)} = \frac{(x-0.6)(x-1.7)}{(0.9)(-0.2)} = \frac{(x-0.6)(x-1.7)}{-0.18}$$

$$L_2(x) = \frac{(x-0.6)(x-1.5)}{(1.7-0.6)(1.7-1.5)} = \frac{(x-0.6)(x-1.5)}{(1.1)(0.2)} = \frac{(x-0.6)(x-1.5)}{0.22}$$

$$p_2(x) = 0.38 \cdot L_0(x) + 1.50 \cdot L_1(x) + 1.49 \cdot L_2(x)$$

Degree 3 (nodes x_0, x_1, x_2, x_3) — showing L_0 as example

$$L_0(x) = \frac{(x-1.5)(x-1.7)(x-1.9)}{(0.6-1.5)(0.6-1.7)(0.6-1.9)} = \frac{(x-1.5)(x-1.7)(x-1.9)}{(-0.9)(-1.1)(-1.3)} = \frac{(x-1.5)(x-1.7)(x-1.9)}{-1.287}$$

The full degree-3 interpolating polynomial is then:

$$p_3(x) = 0.38 \cdot L_0(x) + 1.50 \cdot L_1(x) + 1.49 \cdot L_2(x) + 1.12 \cdot L_3(x)$$

(iii) divided differences.

Step 1 — Divided-Difference Table

i	x_i	$f[\cdot]$	$f[\cdot, \cdot]$	$f[\cdot, \cdot, \cdot]$	$f[\cdot, \cdot, \cdot, \cdot]$
0	0.6	0.38	1.2444	-1.1767	-2.5564
1	1.5	1.50	-0.0500	-4.5000	
2	1.7	1.49	-1.8500		
3	1.9	1.12			

Step 2 — Computing the Entries

The general recurrence relation is:

$$f[x_i, \dots, x_j] = \frac{f[x_{i+1}, \dots, x_j] - f[x_i, \dots, x_{j-1}]}{x_j - x_i}$$

First order:

$$f[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0} = \frac{1.50 - 0.38}{1.5 - 0.6} = \frac{1.12}{0.9} = 1.2444$$

$$f[x_1, x_2] = \frac{1.49 - 1.50}{1.7 - 1.5} = \frac{-0.01}{0.2} = -0.0500 \quad f[x_2, x_3] = \frac{1.12 - 1.49}{1.9 - 1.7} = \frac{-0.37}{0.2} = -1.8500$$

Second order:

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{-0.0500 - 1.2444}{1.7 - 0.6} = \frac{-1.2944}{1.1} = -1.1767$$

$$f[x_1, x_2, x_3] = \frac{-1.8500 - (-0.0500)}{1.9 - 1.5} = \frac{-1.8000}{0.4} = -4.5000$$

Third order:

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = \frac{-4.5000 - (-1.1767)}{1.9 - 0.6} = \frac{-3.3233}{1.3} = -2.5564$$

The Newton coefficients are the top diagonal of the table:

$$a_0 = 0.38, \quad a_1 = 1.2444, \quad a_2 = -1.1767, \quad a_3 = -2.5564$$

Step 3 — Nested Form Polynomials

$$p_1(x) = 0.38 + (x - 0.6) \cdot 1.2444$$

$$p_2(x) = 0.38 + (x - 0.6)[1.2444 + (x - 1.5)(-1.1767)]$$

$$p_3(x) = 0.38 + (x - 0.6)[1.2444 + (x - 1.5)[-1.1767 + (x - 1.7)(-2.5564)]]$$

Summary

All three methods produce the same polynomial at each degree, since by the **uniqueness theorem** there is exactly one polynomial of degree $\leq n$ passing through $n + 1$ distinct data points. Small numerical differences between methods are due to rounding only.

Degree	Newton (nested)	Lagrange	Neville
1	$p_1(x)$	$p_1(x)$	$p_{01}(x)$
2	$p_2(x)$	$p_2(x)$	$p_{012}(x)$
3	$p_3(x)$	$p_3(x)$	$p_{0123}(x)$

H8. Compute nested interpolating polynomials of degree 3 for the data $y = f(x)$ below using divided differences, and approximate the missing value:

a.

x	2.1	2.3	2.6	2.7	2.4
y	6.9	7.6	8.5	8.8	

b.

x	0.25	0.5	0.75	1.0	0.8
y	-0.07	-0.03	0.34	1.10	

Suppose $f(1.1) = 2.00$ is added to the data for (b). Construct the interpolating polynomial of degree 4, and compute a new approximation for the missing value.

Answer

a. Divided differences

i	x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, \dots, x_{i+3}]$
0	2.1	6.9	3.5	-1.0	1.666
1	2.3	7.6	3.0	0.0	
2	2.6	8.5	3.0		
3	2.7	8.8			

Interpolating polynomial

$$\begin{aligned}
 p(x) &= a_0 + (x - x_0)(a_1 + (x - x_1)(a_2 + (x - x_2)a_3)) \\
 &= 6.9 + (x - 2.1)(3.5 + (x - 2.3)(-1.0 + (x - 2.6)1.6)) \\
 p(2.4) &= 7.91.
 \end{aligned}$$

b. Divided differences

i	x_i	$f[x_i]$	$f[x_i, \dots, x_{i+j}]$			
0	0.25	-0.0700	0.1600	2.6400	0.6400	26.5188
1	0.50	-0.0300	1.4800	3.1200	23.1810	
2	0.75	0.3400	3.0400	17.0286		
3	1.00	1.1000	9.0000			
4	1.10	2.0000				

Interpolating polynomials:

$$\begin{aligned}
 p_3(x) &= a_0 + (x - x_0)(a_1 + (x - x_1)(a_2 + (x - x_2)a_3)) \\
 &= -0.070 + (x - 0.25)(0.160 + (x - 0.5)(2.640 + (x - 0.75)0.640)) \\
 p_4(x) &= a_0 + (x - x_0)(a_1 + (x - x_1)(a_2 + (x - x_2)(a_3 + (x - x_4)a_4))) \\
 &= -0.070 + (x - 0.25)(0.160 + (x - 0.5)(2.640 + (x - 0.75)(0.640 + (x - 1.0)26.519))) \\
 p_3(0.8) &= 0.4589 = 0.46 \text{ (2dp)} \\
 p_4(0.8) &= 0.4151 = 0.42 \text{ (2dp)}.
 \end{aligned}$$

- C9.** Write a Matlab function to compute the coefficients a_i of the nested polynomial interpolating data (x_i, y_i) using divided differences.

Write another function to evaluate the polynomial at the point w given the interpolation points x_i and coefficients a_i .

Use your code to compute the interpolating polynomial for the data below, and estimate the values of y for the given x .

a.	x	0.1	0.2	0.3	1.0	0.25		
	y	0.620500	-0.283987	0.006601	0.248424			
b.	x	0.4	0.5	0.6	0.7	0.8	1.0	0.9
	y	-0.36770	-0.19481	-0.17695	-0.01375	0.22363	0.65809	

- H10.** Consider the following functions, interpolation points x_i and unknown x -values.

- a.** $f(x) = x \log x$, $x_i = \{0.3, 0.5, 0.6\}$, $x = 0.4, 0.7$.
b. $f(x) = \cos(3e^x)$, $x_i = \{0.6, 0.7, 0.8, 0.9, 1.0\}$, $x = 0.55, 0.75, 0.85$.

For each case, construct the polynomial interpolants. Use your interpolants to approximate f and f' at the value x given, and calculate the actual error. Which approximations are more accurate, and why?

Answer

- a.** $p(x) = -0.36119 + (x - 0.3) \times (0.07309 + (x - 0.5) \times 1.09230)$.
 $p(0.4) = -0.3648$ (4dp), $e(0.4) = 1.7 \times 10^{-3}$.
 $p(0.7) = -0.2446$ (4dp), $e(0.7) = 5.1 \times 10^{-3}$.
b. $p(x) = 0.685 + (x - 0.6) \times (2.863 + (x - 0.7) \times (-16.681 + (x - 0.8) \times (-14.200 + (x - 0.9) \times 90.122)))$. $p(0.55) = 0.502$, $e(0.55) = 0.034$; $p(0.75) = 0.9993$, $e(0.65) = 0.0016$; $p(0.85) = 0.7398$, $e(0.85) = 0.0016$.

Errors are smaller inside the interval (interpolation) than outside (extrapolation).

- A11.** For the interpolation in Question 10(a), use the error formula for polynomial interpolation to give bounds for the errors of your approximations for $f(x)$, and compare with the actual errors.
- E12.** Approximate $\sqrt{5}$ by interpolating the function $f(x) = 5^x$ at the values $x_0 = -2$, $x_1 = -1$, $x_2 = 0$, $x_3 = 1$ and $x_4 = 2$.
- C13.** Use the Matlab command `sd=spline(xs,ys)` to compute the data for the spline s interpolating the data of Question 4. Examine the data structure `s`, especially the fields `sd.breaks` and `sd.coefs`. Write down the cubic polynomial s_1 defining s on the interval $[0.6, 1.5]$, and compute $s_1(1.0)$.
Check your answer by using the Matlab command `ppval(sd,x)` to evaluate the piecewise-polynomial spline s at the interpolation points, and at $x = 1.0$. Plot a graph of s and compare your results to the polynomial interpolation. Which do you think is more accurate?
- C14.** Recall that for the interval $[-1, +1]$, equally-spaced nodes $x_0, \dots, x_n$ are given by $x_i = 2i/n - 1$, and Chebyshev nodes by $x_i = -\cos((2i+1)\pi/(2n+2))$.

For each of

- (i) Equally-spaced nodes and polynomial interpolation.
- (ii) Chebyshev nodes and polynomial interpolation.
- (iii) Equally-spaced nodes and spline interpolation.

and for $n = 8, 16, 32$, compute and plot the interpolants for the following functions and intervals:

- a.** For $k = n/4, n/2, n$, the Lagrange basis functions interpolating $y_k = 1$, $y_i = 0$ for $k \neq i$
b. Random data y_i taking values in $[-1, +1]$.

- c.** The Gaussian function $f(x) = e^{-x^2}$.
- d.** The function $f(x) = e^{-x^2}$ perturbed by random data taking values in $[-0.01, +0.01]$
- e.** The function $f(x) = e^{-(5x)^2}$.

Comment on your findings.